

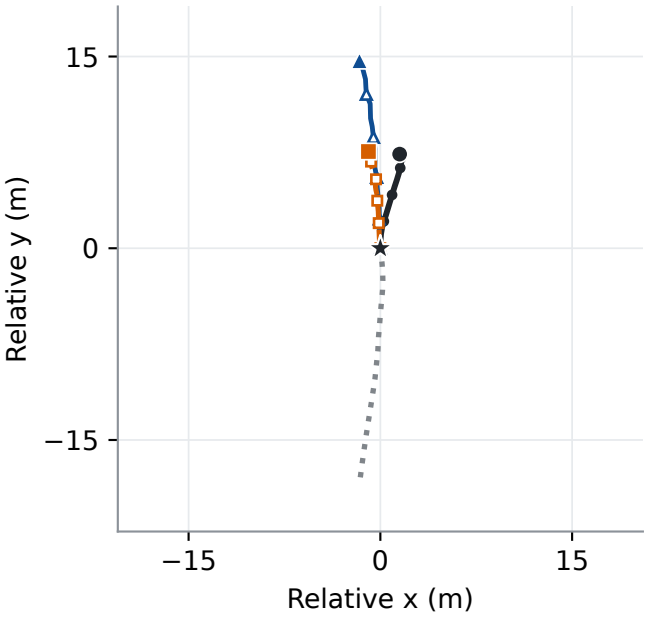
Where inherited QGNN underperforms the strong GNN

Selected cases | V_select, SNR 20 dB | QGNN checkpoint 75; GNN checkpoint 97

..... Past estimate —●— Future truth —▲— QGNN —■— GNN ★ Present target

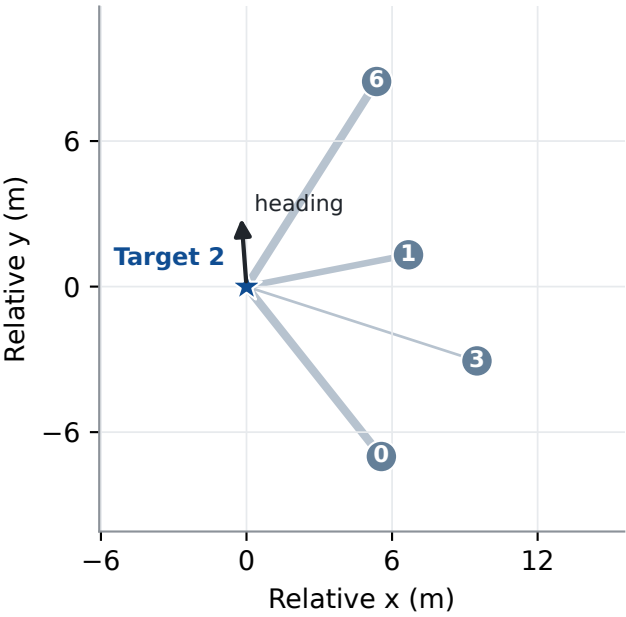
TARGET TRAJECTORY

Case 353 | target slot 2
ADE Q 3.16 / G 1.54 m; FDE Q 7.96 / G 2.45 m



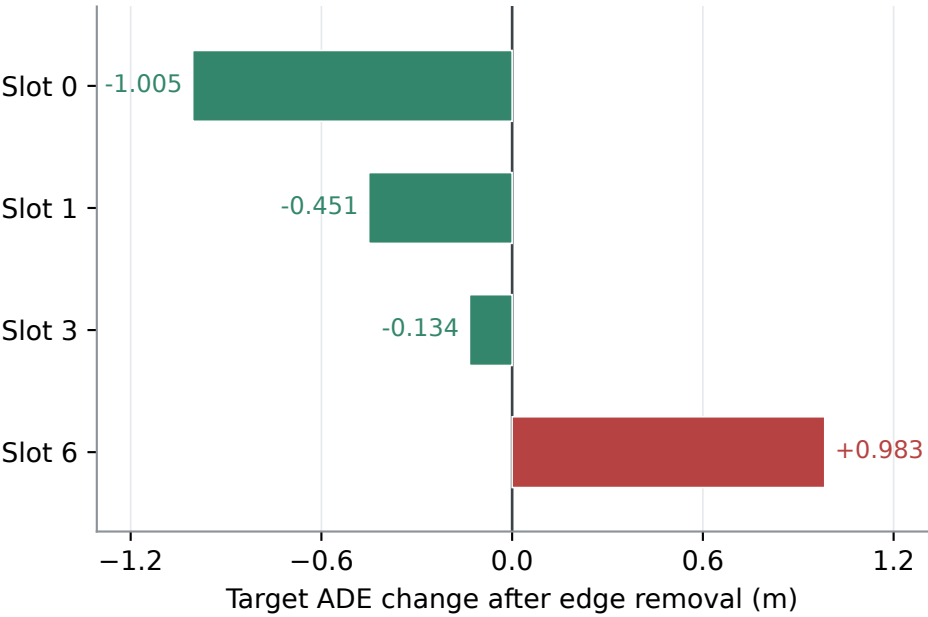
NEIGHBOR CONTEXT

Direct neighbors seen in history
Positions now; width = layer-2 contribution proxy

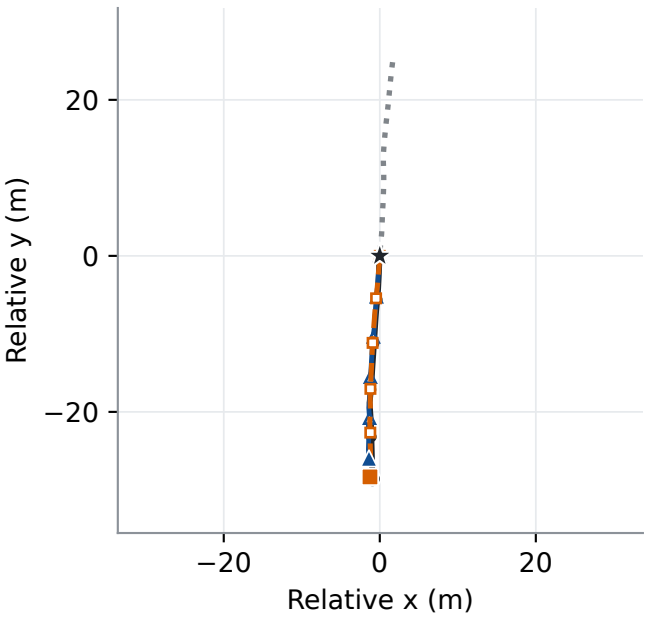


FIXED-MODEL INTERVENTION

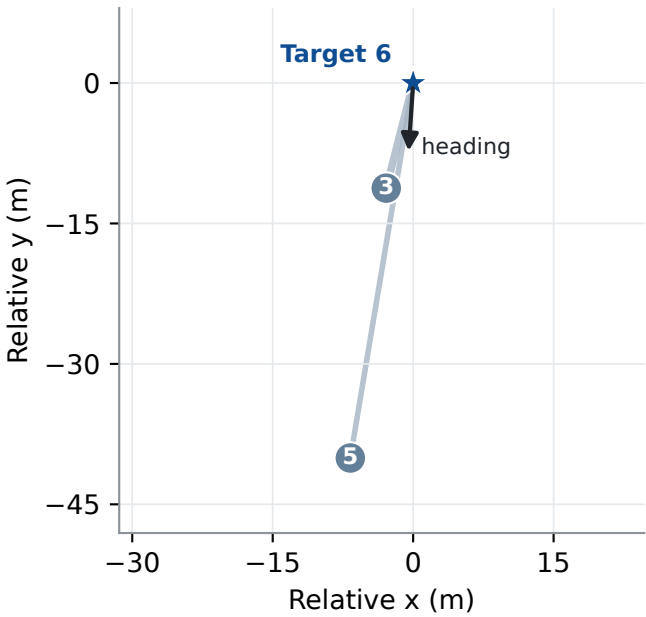
Remove one neighbor-to-target edge
Fixed QGNN; no retraining



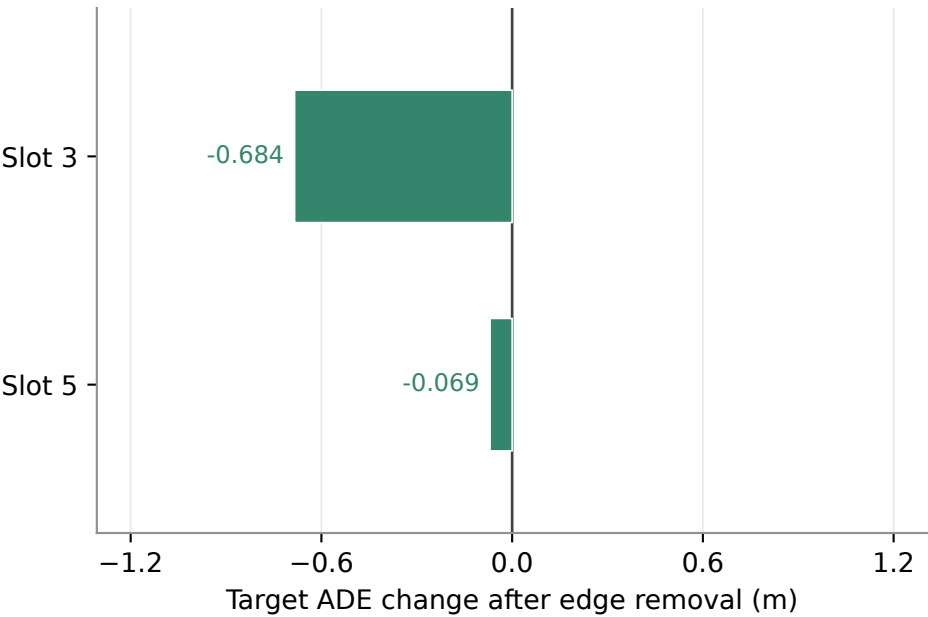
Case 285 | target slot 6
ADE Q 1.35 / G 0.28 m; FDE Q 2.66 / G 0.41 m



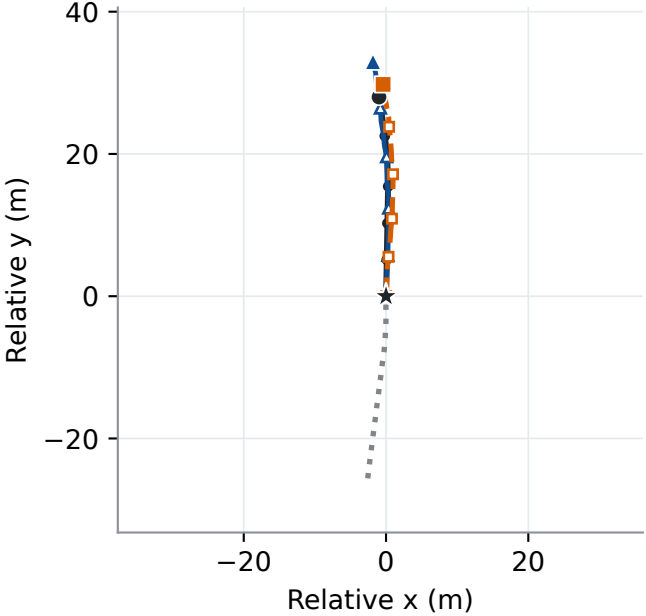
Direct neighbors seen in history
Positions now; width = layer-2 contribution proxy



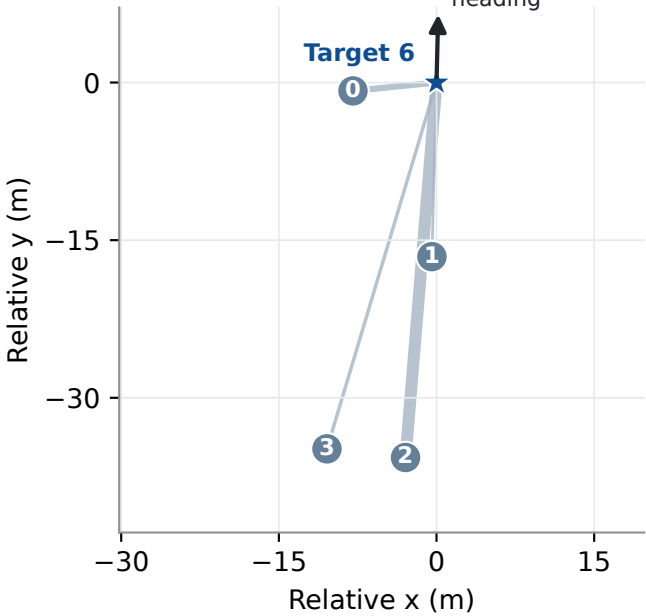
Remove one neighbor-to-target edge
Fixed QGNN; no retraining



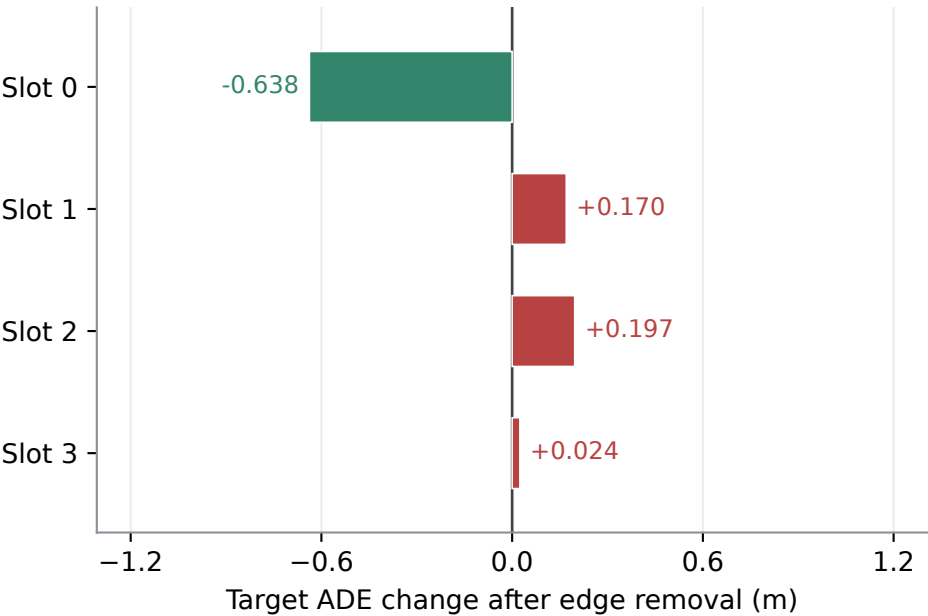
Case 94 | target slot 6
ADE Q 2.72 / G 1.12 m; FDE Q 5.05 / G 1.87 m



Direct neighbors seen in history
Positions now; width = layer-2 contribution proxy



Remove one neighbor-to-target edge
Fixed QGNN; no retraining



Coordinates are translated to the present target; spatial axes have equal meter scales. Neighbor line widths encode history-mean layer-2 contribution norms (proxy only).

Green / negative: removal improves target ADE. Red / positive: removal worsens it. Fixed-model edge removal is a sensitivity diagnostic, not evidence of real-world causal importance.